Aryaman Raj Jaiswal

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Education

Netaji Subhas University of Technology — Dwarka, Delhi
Bachelor of Technology in Electronics and Communications — *Aug. 2024 – 2028*

Experience

SmartOut Media (Canada) — Jan. 2026 – Present
iOS Developer (Contract) — *Remote*

- Rebuilt the SmartOut iOS app from scratch in **Swift/SwiftUI** for a **50K+ install** product: an **offline-first** app that shows Ontario hunters and anglers the regulations for the zone they are in.
- Engineered the map layer with **Google Maps SDK** and **GEOSwift** over **456 regulatory polygons** (MULTIPOLYGON, interior holes): clustering, **polygon hit-testing** to resolve the tapped zone, and **deep links** restoring the exact zone, tab, and row.
- Kept every regulation available offline through a bundled **SQLite** dataset and a **GRDB** store with versioned migrations; traced live issues from **Crashlytics** through the network layer to the query and fixed them end to end.

Bharti Airtel Limited — June 2026 – Aug. 2026
Software Engineer Intern — *Gurugram, India*

- Built a **read-only text-to-SQL agent** (**LangChain**, **NeMo Guardrails**) that lets on-call engineers query the production VM database in English; cut dashboard query latency by **30%** with composite indexing and cursor pagination.

Examination Cell, NSUT — May 2025 – Aug. 2025
Software Developer Intern — *Delhi, India*

- Shipped **NSUTrack**, native **iOS** and Android apps bringing schedules, attendance, and notices to **1,500+ students** with an offline cache and **FCM** push, alongside a Next.js/Flask platform for **10,000+ students** that cut manual data entry by **70%**.

Projects

KilnWatch | *Swift, SwiftUI, MapKit, Swift Concurrency, Swift Testing, AWS* — Oct. 2026

- Co-built a system that turns **satellite detections** of brick kilns into **field inspections**. I built the **native inspector app** and integrated it with the AWS backend, the AI agents, and the resident portal.
- Built the **SwiftUI + MapKit** field experience: inspection routes with synced map pins and stop cards, before/after satellite evidence, measured siting checks, an **Ask** tab that shows the agent's tool trace and citations, and Apple Maps handoff.
- Designed for poor connectivity: an **actor-backed outbox** keeps verdicts and photos across relaunches, offline periods, and partial upload failures, and **idempotent submissions** with presigned **S3** uploads reconcile against the shared registry.
- Wrote a typed **Swift API package** that tolerates unknown enums and missing fields and separates network, server, and decoding failures, covered by **103 Swift tests**; added Dynamic Type, Reduce Motion, and accessible verdict submission.

Wandr | *Swift, SwiftUI, Foundation Models, Network.framework, App Intents, Swift Testing* — 2026

- An app that turns a group chat into a budget-aware outing plan through a **seven-stage**, fully **on-device** LLM pipeline.
- **Foundation Models** return only **indices** into a vetted venue deck under `@Generable` constrained decoding with a **12 s timeout**; code owns every name, price, and time. A **golden set** of 15 requests runs in a **369-test** suite.
- Built group voting on **Network.framework** with **zero dependencies**: `NWConnection` client and `NWListener` relay bridged into **AsyncStream**, 0.25–4 s backoff, a **generation counter** retiring stale callbacks, an offline outbox, and a **pure tally** so all phones agree.

Saldo | *Swift, SwiftUI, Core ML, Vision, VisionKit, Foundation Models, Kotlin* — 2026

- A student finance app that won the **Apple Swift Student Challenge 2026**: **VisionKit OCR** receipts, a **Core ML model I trained**, and the **on-device LLM**, **actor-isolated** with zero cloud or third-party dependencies.
- Savings goals use **Vision subject masking** with a **versioned image pipeline** and automatic migration; later scaled to **Android** with an offline bank-SMS rule engine (13 parser defects fixed, 57 tests).

AquaSense | *Swift, SwiftUI, Speech, AVFoundation, FastAPI, Gemini* — 2026

- Co-built a fish-farm intelligence system, **1st of 200+ teams at MLH Brainwave 2.0**. I built the **SwiftUI** app: an **on-device voice assistant** (Speech transcription, AVSpeechSynthesizer playback), fish-disease photo capture, and a marketplace cart.

Achievements

- **Apple Swift Student Challenge Winner (2026)** for Saldo; **Winner, MLH Brainwave 2.0**: led Team Aquacult to **1st of 200+ teams** (Best Overall, Best Use of Gemini, Best UI/UX Runner-Up).
- **2x Winner, HackonHills (2024, 2025)** among 125+ teams; **HackerRank Orchestrate (2026)**: top 60 of 3,000+.

Technical Skills

Languages: Swift, Python, Kotlin, JavaScript/TypeScript, SQL, C/C++
iOS: SwiftUI, UIKit, Swift Concurrency, Network.framework, App Intents, MapKit, Google Maps SDK, GRDB/SQLite
On-Device AI: Foundation Models (@Generable / @Guide), Core ML, Vision, VisionKit, Speech, AVFoundation
Testing & Tooling: Swift Testing, XCTest, OSLog, Crashlytics, TestFlight & App Store, Xcode, Git, Claude Code
Backend: FastAPI, Flask, REST APIs, PostgreSQL/PostGIS, AWS (Lambda, S3, API Gateway), Firebase